Notes: De Meza and Webb (QJE, 1987)

Too Much Investment: A Problem of Asymmetric Information

Contents

1 One-sentence summary

De Meza and Webb show that asymmetric information in credit markets need not imply credit rationing or underinvestment: when projects differ in expected returns rather than only in risk, pooling debt finance can attract too many low-quality projects, generate excessive aggregate investment, make debt the equilibrium financing contract, and justify an interest-rate tax rather than a subsidy.

2 Big picture

2.1 Question

Can asymmetric information in credit markets lead to *too much* investment rather than too little investment?

2.2 Core answer

Yes. If projects differ in their expected returns because some projects have higher probabilities of success, then pooling debt contracts cause high-quality projects to subsidize low-quality projects. The market-clearing interest rate can finance projects whose expected social return is below the safe return. Thus, equilibrium investment can exceed the first-best level.

2.3 Contrast with Stiglitz-Weiss

Stiglitz and Weiss assume projects have the same expected return but differ in risk. In that setting, higher interest rates attract riskier borrowers, adverse selection can produce credit rationing, and investment tends to be too low. De Meza and Webb change the distributional assumption: projects differ in expected returns. This reverses the sign of the distortion.

2.4 Main policy implication

Under the De Meza-Webb assumptions, a tax on interest income can restore efficiency by raising the cost of bank funds and reducing excessive lending. This is the opposite of the usual credit-rationing policy intuition that subsidized lending is needed.

2.5 Main contracting implication

The assumptions that generate overinvestment also support debt as the equilibrium method of finance. Under the Stiglitz-Weiss assumptions, however, equity finance is the equilibrium contract and removes the underinvestment problem.

3 Incremental contribution of the paper

3.1 Reversing the canonical credit-market intuition

The paper's central contribution is to show that asymmetric information does not have a fixed directional effect on investment. It can generate underinvestment or overinvestment depending on the distribution of project returns.

3.2 From credit rationing to excessive lending

Earlier papers emphasized that adverse selection can ration credit and reduce investment. De Meza and Webb show a different adverse-selection channel: good projects can draw in bad projects, so banks finance socially inefficient marginal borrowers.

3.3 A taxonomy of informational environments

The paper separates two environments that are often conflated:

- **Expected-return heterogeneity:** success probabilities differ and high-probability projects have higher expected returns. This yields overinvestment under debt.
- **Risk heterogeneity with equal expected returns:** safer projects have lower upside but the same expected value. This yields the Stiglitz-Weiss underinvestment logic under debt.

3.4 Contract choice as part of the equilibrium

The paper does not treat debt as an exogenous contract. It asks which financing method survives equilibrium selection. Debt is the equilibrium contract in the overinvestment model, while equity is the equilibrium contract in the Stiglitz-Weiss model.

3.5 Policy contribution

The paper demonstrates that welfare policy depends on the source of asymmetric information. If poor projects are subsidized by good projects, subsidizing credit worsens the distortion; taxing interest can improve efficiency.

3.6 Why the contribution still matters

Modern discussions of small-business finance, credit subsidies, development lending, student loans, and venture finance often presume that credit constraints imply too little investment. De Meza and Webb provide the clean theoretical warning: credit-market frictions can also produce excessive entry and overinvestment.

4 Model setup and measurement

4.1 Agents and timing

There is a continuum of entrepreneurs. Each entrepreneur owns one project and has wealth W. Each project requires the same investment K, with $W < K$, so external finance of size

$$B = K - W$$

is required.

Banks are competitive, risk-neutral lenders. They know the distribution of project characteristics but cannot observe a particular entrepreneur's project quality.

4.2 Returns in the De Meza-Webb model

Every project has the same two possible outcomes:

$$R_s > R_f > 0.$$

Projects differ only in their probability of success p_i. A higher p_i means a better project and a higher expected return:

$$E[R_i] = p_i R_s + (1 - p_i) R_f.$$

Interpretation: This is the key assumption that reverses the Stiglitz-Weiss conclusion. Lower-p_i borrowers are not merely riskier; they also have lower expected returns.

4.3 Debt contract

The entrepreneur borrows B at interest rate r and promises repayment

$$D = (1 + r)B.$$

The paper assumes

$$R_s > D > R_f,$$

so repayment occurs only in the success state and bankruptcy occurs in the failure state.

The entrepreneur's payoff is

$$\pi_i = \max\{R_i - (1 + r)B, 0\}.$$

Expected entrepreneurial payoff under debt is

$$E\pi_i = p_i \left(R_s - (1 + r)B\right).$$

4.4 Entrepreneur participation condition

An entrepreneur undertakes the project if

$$E\pi_i \geq (1 + \rho)W,$$

where ρ is the safe rate. The marginal entrepreneur $\bar{p}$ satisfies

$$\bar{p} \left(R_s - (1 + r)B\right) = (1 + \rho)W.$$

Interpretation: Lower interest rates make debt more attractive and admit lower-p projects. Because these low-p projects also have low expected social returns, low rates can produce overinvestment.

4.5 Bank zero-profit condition

For projects with success probability above the marginal level, expected bank profit is

$$E\Pi^B = (1 + r)B \int_{\bar{p}}^{1} pf(p)dp + R_f \int_{\bar{p}}^{1} (1 - p)f(p)dp - (1 + \rho)B.$$

In competitive equilibrium banks earn zero expected profit on the pool.

5 Model I: overinvestment with expected-return heterogeneity

5.1 Market clearing

The paper first shows that a rationing equilibrium cannot occur in this model. If some profitable borrowers are rationed, a bank can slightly increase the interest rate and lend to the remaining rationed borrowers. The higher rate both raises repayment in success states and improves the average pool because marginal low-p borrowers drop out.

5.2 First-best investment rule

A project should be financed if its expected return exceeds the safe return on the total investment:

$$p_i R_s + (1 - p_i) R_f \geq (1 + \rho) K.$$

5.3 Why competitive lending overinvests

At the market equilibrium, the marginal entrepreneur is willing to borrow because her private upside is positive. But limited liability means she does not fully internalize the downside failure payoff R_f that goes to lenders. High-p borrowers subsidize low-p borrowers in the pooled debt contract, allowing some projects with socially low expected returns to receive funding.

5.4 Proposition 1: the credit market clears

Result. In the De Meza-Webb model, equilibrium must be market clearing.

Interpretation: This is important because the paper's overinvestment result is not caused by rationing. It arises even in a competitive market that clears.

5.5 Proposition 2A: equilibrium investment exceeds first best

Result. If the supply of funds to the banking system is not decreasing in the deposit rate, competitive equilibrium investment exceeds the first-best level.

Economic message: The marginal privately financed project is too low quality from a social perspective. Pooled debt contracts create excessive entry.

5.6 Proposition 2B: backward-bending funds supply

If the supply of funds is backward-bending, equilibrium investment can fall short of the efficient level. This caveat clarifies that the overinvestment conclusion relies on the standard upward-sloping or flat supply of funds condition.

5.7 Proposition 3: interest tax restores efficiency

Result. A tax on interest income can support the first-best allocation.

Interpretation: The tax raises the gross return banks must pay depositors and reduces lending to socially excessive marginal projects. This is the opposite of a subsidized-credit policy.

6 Comparison with the Stiglitz-Weiss model

6.1 Stiglitz-Weiss return structure

In the Stiglitz-Weiss environment, projects differ in risk but have the same expected return:

$$p_i R_i^s + (1 - p_i) R_f = \text{constant}.$$

Riskier projects have lower p_i but higher upside R_i^s.

Interpretation: In this environment, a higher interest rate attracts borrowers with riskier projects because they value the upside more. This is the usual adverse-selection channel.

6.2 Proposition 4: credit rationing may exist

Result. A credit-rationing equilibrium may exist in the Stiglitz-Weiss model.

The reason is that raising the interest rate has two effects:

- it raises repayment from successful borrowers;
- it worsens the borrower pool by attracting riskier entrepreneurs.

If the second effect dominates, banks do not raise rates to clear the market.

6.3 Proposition 5A: underinvestment under debt

Result. If the supply of funds is not decreasing in the deposit rate, equilibrium investment in the Stiglitz-Weiss debt model is below the first-best level.

Interpretation: Under debt, safer projects subsidize riskier projects. The marginal borrowers at the safe end drop out as interest rates rise, causing underinvestment.

6.4 Proposition 7 and Corollary: rationing can mitigate underinvestment

De Meza and Webb show that if the deposit rate rises with bank borrowing, investment is higher under a rationing equilibrium than under the market-clearing rate. Thus, prohibiting rationing can reduce welfare in the Stiglitz-Weiss model.

Economic message: This is a subtle point: credit rationing is not necessarily the source of underinvestment; it may partially offset it relative to market clearing.

7 Contract choice

7.1 Why maximum self-finance is required

Entrepreneurs with better-than-average projects prefer to finance as much as possible internally because outside finance involves pooling and cross-subsidies. Banks infer that borrowers who do not invest their own wealth are worse than average, so maximum self-finance is part of equilibrium contracts.

7.2 Debt versus equity in the De Meza-Webb model

With expected-return heterogeneity, debt attracts higher-success-probability entrepreneurs relative to equity. Equity is less attractive to good borrowers because it requires giving up a proportional share of both success and failure payoffs.

7.3 Proposition 8: debt is the equilibrium contract

Result. In the De Meza-Webb model, equilibrium requires all firms to be debt financed.

Interpretation: Debt is not assumed; it is selected by the asymmetric-information environment. The same assumptions that generate overinvestment also make debt the equilibrium method of finance.

7.4 Equity in the Stiglitz-Weiss model

When all projects have the same expected return but differ in risk, equity finance eliminates the adverse-selection problem because all projects are equally attractive to equity investors in expected value.

7.5 Proposition 9: equity is the equilibrium contract in Stiglitz-Weiss

Result. In the Stiglitz-Weiss model, the equilibrium method of finance is equity, and with all-equity finance the first-best allocation obtains.

Interpretation: This is a sharp critique of treating debt as exogenous in the credit-rationing model. If equity is feasible and moral hazard is absent, the Stiglitz-Weiss underinvestment problem disappears.

8 Identification-style interpretation

8.1 What is being isolated

The paper is theoretical. Its identification-style logic is comparative mechanism design: change the distribution of project returns while holding the asymmetric-information setup fixed, and observe how equilibrium investment and contract choice change.

8.2 Key comparative statics

- Expected-return heterogeneity + debt pooling $\Rightarrow$ overinvestment.
- Risk heterogeneity + debt pooling $\Rightarrow$ adverse selection and underinvestment.
- Expected-return heterogeneity $\Rightarrow$ debt is selected.
- Equal-expected-return risk heterogeneity $\Rightarrow$ equity is selected.

8.3 Why the comparison is powerful

The paper uses almost the same credit-market primitives as Stiglitz-Weiss, but changes how project returns differ across entrepreneurs. This isolates the fact that asymmetric information alone does not determine the direction of inefficiency.

8.4 Main limitations

- Entrepreneurs and banks are risk neutral.
- Moral hazard is absent.
- Projects have simple two-state returns.
- Banks know the population distribution but not individual project types.
- The model compares pure debt and pure equity contracts, though it argues the restriction does not affect the substance.

9 Tables and figures

9.1 Notes-created Tables A and B: Environment and equilibrium direction

Object	De Meza-Webb model
Project returns	Same R_s, R_f across projects
Heterogeneity	Success probability p_i
Expected return	Higher p_i means higher expected return
Debt pool subsidy	Good projects subsidize bad projects
Investment bias	Too much investment
Policy	Interest-income tax
Contract	Debt equilibrium

(a) Table A: De Meza-Webb environment.

Object	Stiglitz-Weiss model
Project returns	Riskier projects have higher upside
Heterogeneity	Risk holding expected return fixed
Expected return	Same across projects
Debt pool subsidy	Safe projects subsidize risky projects
Investment bias	Too little investment under debt
Policy	Interest subsidy if debt is exogenous
Contract	Equity equilibrium if feasible

(b) Table B: Stiglitz-Weiss comparison.

Notes-created comparison tables. The two tables show why changing the distribution of project returns reverses the direction of the credit-market distortion.

Read: Tables A and B show that the key difference is not whether information is asymmetric, but how private information is distributed across projects.

Economic message: The paper's strongest message is that asymmetric information has no universal policy sign. Credit subsidies are desirable in one environment and harmful in the other.

9.2 Notes-created Tables C and D: Propositions and policy mapping

Result	Content
P1	De Meza-Webb equilibrium clears.
P2A	Investment exceeds first best under normal funds supply.
P2B	Backward-bending funds supply can reverse this.
P3	Interest tax can restore efficiency.
P4	Stiglitz-Weiss rationing can exist.
P5A	Debt-financed Stiglitz-Weiss investment is too low.
P7	Rationing can yield more investment than market clearing.
P8	Debt is equilibrium finance in De Meza-Webb.
P9	Equity is equilibrium finance in Stiglitz-Weiss.

(a) Table C: Proposition map.

Diagnosis	Policy implication
Good projects draw in bad	Raise borrowing cost / tax interest
Bad projects drive out good	Subsidize borrowing if debt is fixed
Equity feasible in SW	No subsidy needed; equity restores efficiency
Debt selected in DMW	Overinvestment is equilibrium-relevant
Credit rationing in SW	May reduce, not cause, inefficiency

(b) Table D: Policy map.

Notes-created proposition and policy tables. These are not original tables from the paper; they summarize the theoretical architecture.

Read: Table C organizes the sequence of formal results. Table D maps each theoretical diagnosis into the policy conclusion.

Interpretation: The two tables show that the paper is built as a sequence of reversals: overinvestment versus underinvestment, tax versus subsidy, debt versus equity, and market clearing versus rationing.

9.3 Single summary table: direction of distortions

Environment	Debt equilibrium distortion	Efficient response
Expected-return heterogeneity	Too much investment	Tax interest income
Equal expected returns, risk heterogeneity	Too little investment if debt is imposed	Equity or interest subsidy under debt
Equity feasible in SW	No adverse selection over expected returns	First-best possible

Single unpaired notes-created table. This unpaired table is set to `width=0.6`
`linewidth` through the enclosing minipage, following the requested single-visual layout rule. It summarizes the sign of the distortion.

10 Short summary

- De Meza and Webb study credit markets with asymmetric information about project quality.
- Their key model assumes projects share the same success and failure payoffs but differ in success probabilities.
- This means projects differ in expected returns, not only in risk.
- Pooling debt contracts allow low-quality projects to be cross-subsidized by high-quality projects.
- Competitive equilibrium can therefore feature too much investment.
- An interest-income tax can restore efficiency.
- The paper contrasts this with Stiglitz-Weiss, where projects have the same expected return but differ in risk.
- Under Stiglitz-Weiss assumptions, debt generates adverse selection and underinvestment.
- However, if equity is feasible, the Stiglitz-Weiss underinvestment problem disappears.
- Contract choice depends on the informational environment: debt in De Meza-Webb, equity in Stiglitz-Weiss.

11 Conclusion

11.1 Core conclusion

The paper's core conclusion is that asymmetric information in credit markets can generate overinvestment, not only underinvestment. The sign of the distortion depends on whether borrowers differ in expected returns or merely in risk.

11.2 Economic intuition

When success probabilities differ and success/failure payoffs are common, safer borrowers are socially better borrowers. A pooled debt contract makes these good borrowers subsidize worse borrowers, allowing marginal bad projects to be funded. This creates excessive investment. In contrast, when projects have equal expected returns but differ in risk, higher interest rates attract riskier borrowers, generating the standard underinvestment result under debt.

11.3 Contribution revisited

The paper changes the interpretation of adverse selection in credit markets. It shows that "credit-market failure" is not a synonym for too little credit. Whether policy should subsidize or tax borrowing depends on the underlying information problem.

11.4 Implications for modern finance

The result is relevant for credit subsidies, small-business lending, public guarantees, student loans, development finance, and fintech lending. Programs designed to expand credit can reduce welfare if the marginal funded projects have low expected social returns.

11.5 Limitations

The model abstracts from moral hazard, collateral, risk aversion, dynamic relationships, costly screening, renegotiation, and richer securities. It is a clean benchmark rather than a full theory of credit markets.

11.6 Takeaway

The enduring lesson is simple: asymmetric information changes the composition of funded projects. That composition effect can imply either too little or too much investment. Good policy requires knowing which margin dominates.